

ToCoAD: Two-Stage Contrastive Learning for Industrial Anomaly Detection

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Abstract—Unsupervised industrial anomaly detection methods can identify product defects in industrial images using only normal samples for training. Existing approaches perform well on public industrial image datasets but struggle with specific anomaly types due to the domain gap between pretrained feature extractors and target-specific domains. To tackle this issue, this article presents a two-stage training strategy, called ToCoAD. In the first stage, a discriminative network is trained by using synthetic anomalies in a self-supervised learning manner. This network is then utilized in the second stage to provide a negative feature guide, aiding in the training of the feature extractor through bootstrap contrastive learning. This approach enables the model to progressively learn the distribution of anomalies specific to industrial datasets, effectively enhancing its generalizability to various types of anomalies. Extensive experiments are conducted to demonstrate the effectiveness of our proposed two-stage training strategy, and our model produces competitive performance, achieving pixel-level AUROC scores of 98.21%, 98.43%, and 97.70% on MVTec AD, VisA, and BTAD, respectively.

Index Terms—Anomaly detection, contrastive learning, industrial manufacturing, self-supervised learning, synthetic anomaly.

I. INTRODUCTION

INDUSTRIAL anomaly detection aims to identify defective products during quality inspection sessions within the production process, intending to improve the yield rate. Recently, interest in this field has surged due to the growing demands of industrial development [1], [2], [3]. The anomaly detection models employ the image data of the sample, such as battery images captured from a fixed position or screw images at various rotation angles, to determine whether the sample is anomalous or not and to localize the anomalies within the image. Given the challenge of obtaining sufficient anomalous samples, the distribution of anomalies is typically nonestimable, making this scenario classified as an

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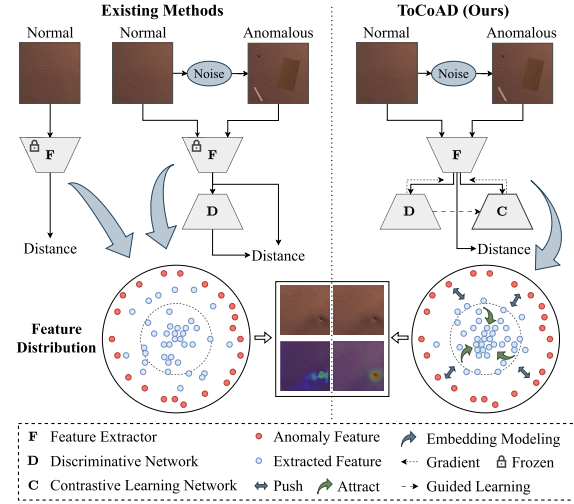


Fig. 1. Existing methods rely on frozen pretrained feature extractors, which can lead to inaccuracies in anomaly detection. In contrast, our method utilizes a two-stage training strategy to fine-tune the feature extractor under the contrastive learning paradigm.

unsupervised learning task. The objective is to train models using only defect-free images, enabling them to detect and localize anomalous regions during the testing phase. Specifically, during the training phase, only normal images are used to train models in an unsupervised learning manner. For example, feature embedding-based [4], [5], [6] methods typically use Gaussian distribution or normalizing flow (NF) [7] to model the distributions of normal samples, representing a normal prototype or feature set. During the test phase, anomalies are identified by comparing the features of the test image with those in the coreset. On the other hand, reconstruction-based [8], [9], [10] methods utilize a reconstruction network to generate an image that closely resembles the normal image during training. Anomalies are then detected by analyzing the differences between the input image and the reconstructed image during testing.

Existing anomaly detection methods, predominantly leveraging deep learning techniques [11], [12], have shown superior performance in industrial benchmarks [13]. As illustrated in Fig. 1, the majority of existing methods, including feature embedding-based [4], [5], [6] and synthetic anomaly-based methods [14], [15], [16], have attempted to train a classifier or discriminative network to identify anomalies by utilizing a pretrained truncated model for extracting features. However, it is challenging to accurately identify all anomalies,

primarily due to the indistinguishability of frozen pretrained models from normal and anomalous features. This issue arises from the domain gaps [17] between pretrained datasets (e.g., ImageNet [18]) and target-specific domains (i.e., industrial images). Specifically, the anomaly detection task is limited by two factors. First, the covariate shift phenomenon, caused by the domain gap between source and target data, can hinder anomaly localization [19], making it difficult for a pretrained network to extract discriminative features for anomalies. Second, the features learned during pretraining are typically optimized for different tasks, but they may not be fully suitable due to the imbalance between normal and anomalous samples [20]. For example, semantic segmentation and image classification often use balanced datasets with predefined labels, whereas anomaly detection is formulated in an unsupervised learning manner, and data deviating from the normal distribution is treated as anomalous. Recently, reconstruction-based methods [8], [9] have been developed to train a decoder and feature extractor jointly by learning an identical map between the input and output. However, the performance of these methods remains suboptimal, limited by the training overhead and the ability to reconstruct large areas of anomalies. Recently, the application of contrastive learning has demonstrated considerable progress in cross-domain learning [21]. Fine-tuning the network using positive sample pairs or negative samples has been demonstrated to be effective in enhancing the network's generalizability to domain-specific datasets, indicating significant potential in the field of anomaly detection as well. Contrastive learning has recently proven effective for label-efficient fine-tuning [20], [22]. It has the potential to mitigate domain gaps and help the network adapt to the target domain data.

In order to mitigate the domain gaps between source and target data, and to fully utilize the advantages of contrastive learning in unsupervised fine-tuning training, we propose a two-stage training strategy that incorporates a contrastive learning framework to adapt to the target domain data by learning robust features from target samples through self-supervised learning [23]. Specifically, we initially leverage a frozen feature extractor to obtain generalized features and train a discriminative network progressively to identify anomalies using synthetic anomalous samples. In the second stage, the discriminate network is fixed to provide a negative guide, while a bootstrap contrastive learning is employed to fine-tune the feature extractor and contrastive learning network jointly. This joint learning can be viewed as a form of adversarial learning [24] of defect and defect-free features, where normal features are compactly enclosed and distinct from defective features. Finally, the fine-tuned feature extractor is used to extract patch features to construct a memory bank through coreset subsampling, and distance metrics between the test sample features and the memory bank features are computed to localize anomalies during the inference phase.

Our contributions can be summarized as follows.

- 1) We propose a two-stage training strategy to fine-tune the feature extractor to bridge the domain gap between pretrained and target features. In the first stage, a network is progressively trained to localize anomalies coarsely,

and then it is employed to facilitate the fine-tuning of the extractor in the second stage.

- 2) We introduce a negative-guided contrastive learning (NCL) paradigm, which utilizes the discriminative network guiding negative bootstrap contrastive learning to fine-tune the feature extractor. A contrastive loss is introduced to regularize negative bootstrap joint learning.
- 3) Our model shows competitive performance on three popular anomaly detection datasets. It achieves AUROC scores of 99.10%/98.21% (image-/pixel-level) on MVTec AD, 95.35%/98.43% on VisA, and 97.70% (pixel-level) on BTAD.

II. RELATED WORK

A. Anomaly Detection

Most anomaly detection methods utilize a feature extractor pretrained on ImageNet [18] to obtain generalized features. These methods can be classified into three types: feature embedding-based, reconstruction-based, and synthetic anomaly-based methods.

1) *Feature Embedding-Based Methods*: They typically utilize pretrained models to extract features and directly model the distributions using various machine learning techniques. For example, PaDiM [5] decomposes the image into patches to obtain a probabilistic representation of the normal class using multivariate Gaussian distributions. PatchCore [4] constructs a memory bank for storing neighborhood-aware patch-level features. Some methods are based on NF [7] to convert pretrained feature distributions into simple distributions. CFLOW-AD [6] utilizes a pretrained encoder with multi-scale pyramid pooling to capture rich features and leverages conditional normalization flows to enhance anomaly detection efficiency. PyramidFlow [25] utilizes invertible pyramids and coupled pyramid blocks to localize anomalies through multiscale feature interaction. These methods use pretrained feature extractors, such as ResNet [26], WideResNet [27], and EfficientNet [28] to extract features for modeling distributions.

2) *Reconstruction-Based Methods*: They assume that models trained on only normal samples cannot accurately reconstruct anomalous regions, thus allowing for the localization of anomalies by identifying differences between the input and reconstruction results. For instance, RIAD [9] and InTra [8] perform mask operations on normal samples, and train reconstruction models to recover the masked regions. DiffusionAD [29] uses a diffusion model [30] to reconstruct normal samples as near-normal samples, and localizes anomalous regions using a segmentation network.

3) *Synthetic Anomaly-Based Methods*: They formulate unsupervised anomaly detection as a binary classification task, in which pseudo anomaly samples are generated to train the discriminative network for identifying anomalies. CutPaste [16] employs an augmentation strategy where a smaller patch is replaced by other regions. DRAEM [14] combines Perlin noise [31] and binarized masks of samples to generate anomalous images for training their discriminative networks. NSA [15] employs Poisson image editing techniques to achieve a seamless fusion of anomalies.

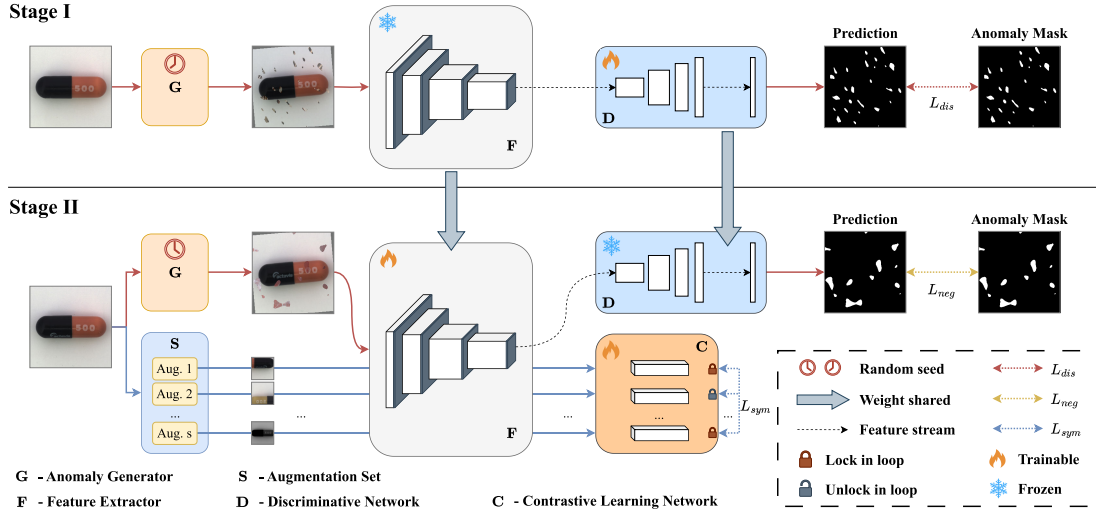


Fig. 2. Overview of ToCoAD. First, a synthetic anomaly image is generated by anomaly generator G to train a discriminative network D . Then, the feature extractor F is fine-tuned jointly by contrastive learning network C and pretrained D using synthetic anomaly images and augmented image pairs.

However, most of these methods [4], [5], [6], [32], [33], [34], [35] rely on feature extractors pretrained on ImageNet [18], leading to a domain gap between pretrained features and industrial target features. To alleviate this problem, our ToCoAD employs a two-stage training approach that uses positive and negative samples to jointly fine-tune the feature extractor, guiding it to acquire adaptive feature representation capabilities.

B. Contrastive Learning

Recently, contrastive learning [36], [37], [38] has played a significant role in self-supervised learning, offering a promising paradigm for exploiting unlabeled data without the need for human annotations. The concept aims to enhance feature consistency and obtain invariant feature representations between different views of the same sample while preserving the differences among other samples. For example, SimCLR [39] constructs positive pairs from the same sample and negative pairs from different samples, employing an N-pair loss [40] to keep positive pairs close and negative pairs far away. BYOL [37] avoids prediction crashes and removes negative pairs by an exponential moving average model. SimSiam [36] addresses the issue of collapsing solutions without using negative samples by employing a stop-gradient (SG) operation and a predictor.

Given the capability of contrastive learning methods to enable models to acquire robust and generalized feature representations [41], we propose a negative bootstrap contrastive learning to fine-tune the feature extractor. Our approach uses augmented positive samples to train the model by learning feature representations of normal samples while simultaneously using synthetic anomaly samples to bootstrap the model, ensuring its sensitivity to anomalous features.

III. METHOD

Given a training set $\mathcal{T}_{\text{train}}$ comprising N images and a test set $\mathcal{T}_{\text{test}}$ consisting of N' images, each image $I \in \mathbb{R}^{H \times W \times C}$ corresponds to a binary label $c = \{0, 1\}$ where H , W , and

C denote the height, width, and the number of the channel of the image. For unsupervised anomaly detection, all images in $\mathcal{T}_{\text{train}}$ are categorized into normal (e.g., defect-free) images with $c = 0$, while $\mathcal{T}_{\text{test}}$ may contain anomalous images with $c = 1$ and a corresponding binary annotation $y \in \{0, 1\}^{H \times W \times C}$ indicating the anomalous regions. The normal images in the training set and the test set come from the same normal data distribution with similar sample characteristics, while the anomalous images deviate from these normal samples. The goal is to classify and localize anomalous regions in $\mathcal{T}_{\text{test}}$.

As shown in Fig. 2, we propose a two-stage training strategy to learn a feature representation adapted to the target data distribution. The first stage is to train a discriminative network to detect anomalies and localize anomalous regions coarsely. It includes an anomaly generator G , a feature extractor F and a discriminative network D . In the second stage, the discriminative network pretrained in stage I is incorporated with the contrastive learning network C , and these networks are jointly trained in a bootstrap contrast learning manner. Finally, a memory bank $\mathcal{M}$ is constructed for storing the normal features, which is used to estimate the anomaly score maps during the testing phase.

A. Discriminative Network Pretraining

Given the unavailability of anomalous samples for training a discriminative network, we initially utilize an anomaly generator G to synthesize pseudo anomalies from the training set. The generation of anomalies can be implemented in various forms by using self-supervised learning techniques [15], [16] or by injecting noises into normal images [14]. To obtain discriminative representations of various artificial defects, we employ widely used Perlin noise [31] as the anomaly generator G , which synthesizes pseudo-anomaly samples by injecting Perlin noise into normal images. Anomalies synthesized using Perlin noise and texture images exhibit irregular geometric traits yet share similar variations with real-world anomalies in terms of color, shape, and location. These similarities allow the model to leverage synthetic anomalies as reference boundaries for learning the boundary outside the

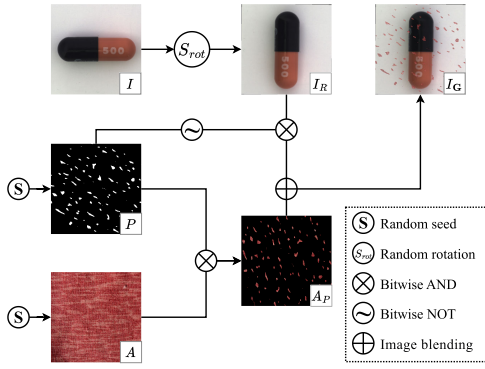


Fig. 3. Overview of anomaly generator $\mathbf{G}$ with Perlin noise. First, the normal image I undergoes a random slight angle and 90° rotation to obtain I_R . Second, the Perlin noise P and the texture sample A are subjected to a bitwise operation to generate the anomalous region A_P . Finally, the anomalous region A_P and the rotated image I_R are fused to obtain the synthetic anomaly sample I_G .

normal data. As a result, the model builds a more compact and refined representation of normal data, effectively distinguishing between normal and anomalous data.

As shown in Fig. 3, random slight angle rotations and $\{0^\circ, 90^\circ, 180^\circ, 270^\circ\}$ rotations are applied to a base (normal) image to enhance normal sample diversity. Perlin noise is then generated using a random seed, with its threshold adjusted to control the size, quantity, and position of the noise. An additional texture dataset, like the Describable Textures Dataset [42], is used as a reference to combine with normal samples based on the Perlin noise mask, creating synthetic anomalous regions. These regions are then pasted onto the base image to form the synthetic anomaly image and corresponding mask. Perlin noise is utilized to generate masks with varying shapes, sizes, and positions. These masks, when combined with random anomalous data, introduce a wide range of texture and color anomalies across different regions of the image. In addition, when applied to the edges of objects, the masks not only introduce visual irregularities but also induce shape distortions, altering the form and outline of the object. A normal image $I \in \mathcal{T}_{train}$ is passed through the anomaly generator $\mathbf{G}$ to produce a pseudo-anomaly sample I_G with an anomaly mask $y_G \in \{0, 1\}^{H \times W \times C}$. A pseudo-anomaly dataset can be obtained

$$\mathcal{T}_G = \mathbf{G}(I_i) \quad \forall I_i \in \mathcal{T}_{train} \quad (1)$$

where $i = \{1, \dots, N\}$ denotes the index in the training set.

Then, each pseudo-anomaly image is used to train the discriminative network $\mathbf{D}$, where the pretrained feature extractor is fixed during the early stages of model training. The architecture of the discriminative network is symmetric with the feature extractor, such as the WideResNet50 [27]. This network operates by successively up-sampling and splicing the extracted features to finally output a feature map $\hat{y}_G$ as a predicted mask with the same dimensions as the input image I_G . The training loss L_{dis} is computed as

$$L_{dis} = L_{focal}(\hat{y}_G, y_G) = -\alpha_t(1 - p_t)^\gamma \log(p_t) \quad (2)$$

where α_t is the scaling factor associated with the category t , γ is an adjustable parameter, and p_t corresponds to the predicted

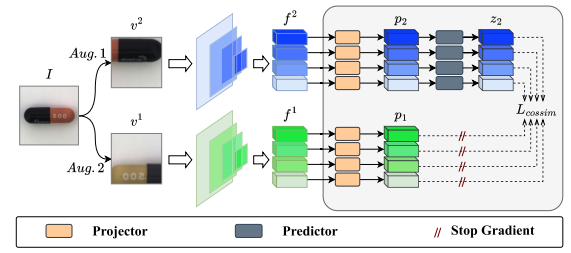


Fig. 4. Detailed structure of contrastive learning network $\mathbf{C}$. A normal image I is augmented to obtain two views v^1, v^2 , which are then passed through the feature extractor and a contrastive learning network to obtain the projected features z_1, z_2 and predicted features p_1, p_2 . The L_{cossim} is calculated for two branches, one of which has no predictor branch applying the SG operation.

pixel point classification with 1 for the anomalous category and 0 for the normal category. The focal loss [43] effectively addresses the sample imbalance problem in pixel-level one-class classification. Upon completion of the discriminative network training, the extracted feature map can coarsely localize the probability of anomalies in the input image.

In contrast to the end-to-end simultaneous training approach [14], [15] and the approach of training the backbone network before training the discriminative network [34], our method first leverages the powerful feature representation capabilities of a pretrained network on ImageNet to train an anomaly discriminative network, fostering coarse-grained localization capability. Subsequently, we use the discriminative network to reversely fine-tune the feature extraction network, encouraging the latent space of normal samples more compact and reducing the domain gap in the feature extraction network.

B. Negative-Guided Contrastive Learning

In stage II, the discriminative network $\mathbf{D}$ is employed to train the feature extractor $\mathbf{F}$ and the contrastive learning network $\mathbf{C}$. Specifically, discriminative network $\mathbf{D}$, pretrained in stage I, acts as a guide for providing negative sample features in bootstrap learning. We employ the advanced SimSiam [36] as the contrastive learning network $\mathbf{C}$ due to its simplicity and exceptional capability in achieving robust and generalized feature representation. As shown in Fig. 4, for training the contrastive learning network $\mathbf{C}$, a normal image I is augmented using a data augmentation set $\mathbf{S}$ (such as random crop, color jitter, and grayscale) to generate M different views $v^1, \dots, v^M$. We begin by randomly cropping the image using a scale factor ranging from 0.2 to 1 to obtain image patches. Next, Gaussian filtering with a standard deviation of $\sigma = 0.1 - 2.0$ is randomly applied to these patches, allowing certain features to emerge in different states. Finally, to reduce the model's reliance on image orientation and mitigate directional bias, we apply a random horizontal flip with a probability of $p = 0.5$ to the patches. These views are fed into the feature extractors $\mathbf{F}$ to extract features from different layers, which are considered as generic feature representations from a normal image, and these features contain both the low-level texture detail feature representation and the high-level abstract semantic feature representation.

For constructing contrastive instance pairs, we take an input image I with two augmented views v^1 and v^2 (M set 2). As an

example, the features f^1 are extracted from v^1 using feature extractor $\mathbf{F}$ and then are passed into the projector and predictor sequentially to obtain z_1 and p_1 , respectively. Then, the cosine similarity loss can be obtained

$$\mathcal{D}(p_1, z_2) = -\frac{p_1}{\|p_1\|_2} \cdot \frac{z_2}{\|z_2\|_2} \quad (3)$$

where $\|\cdot\|_2$ denotes the L_2 distance of an output feature. Following previous work [36], [41], the SG operation is employed to halt the gradient propagation of z_1 and z_2 , effectively preventing collapsing solutions. Since $\mathcal{D}$ is asymmetric, the average bidirectional similarity between f^1 and f^2 is computed to maintain equilibrium, as shown below

$$L_{\text{cosim}}(f^1, f^2) = \frac{1}{2}\mathcal{D}(p_1, \text{SG}(z_2)) + \frac{1}{2}\mathcal{D}(p_2, \text{SG}(z_1)). \quad (4)$$

When generating M augmented samples from each normal sample, the symmetric cosine similarity loss is computed as

$$\begin{aligned} L_{\text{sym}} &= \frac{2}{M(M-1)} \sum_i^M \sum_{j \neq i}^M L_{\text{cosim}}(f^i, f^j) \\ &= \sum_i^M \sum_{j \neq i}^M \frac{[\mathcal{D}(p_j, \text{SG}(z_i)) + \mathcal{D}(p_i, \text{SG}(z_j))]}{M(M-1)}. \end{aligned} \quad (5)$$

For synthetic anomaly sample, we obtain its pseudo-anomaly mask y'_G and predictive feature map $\hat{y}'_G$, and compute the focal loss $L_{\text{neg}} = L_{\text{focal}}(\hat{y}'_G, y'_G)$ as (2). The training loss in the second training stage is shown as

$$L_{\text{ncl}} = \lambda \cdot L_{\text{sym}} + (1 - \lambda) \cdot L_{\text{neg}} \quad (6)$$

we followed the previous methods [14], [34], where λ is a hyperparameter used to balance the two losses.

C. Memory Modeling and Anomaly Detection

During the inference phase, we utilize the feature extractor as a backbone to extract adapted features from various layers. These features are then compressed into a memory bank via the coreset selection [4]. To concurrently harvest shallow texture information and deep semantic content, features extracted from both layers 2 and 3 are aggregated to derive a comprehensive feature for each image. These features are then used to construct the original memory bank $\mathcal{M}_O$. To minimize storage and computation overhead, we use the greedy coreset sampling algorithm to identify the most representative features. This results in a sampled memory bank $\mathcal{M}$ obtained by solving the following equation using iterative greedy approximation suggested in [44]

$$\mathcal{M}^* = \arg \min_{\mathcal{M} \subset \mathcal{M}_O} \max_{m \in \mathcal{M}_O} \min_{n \in \mathcal{M}} \|m - n\|_2. \quad (7)$$

When testing an input image $I_t \in \mathcal{T}_{\text{test}}$, the pixel-level anomaly score s_t is calculated by the maximum Euclidean distance between its adapt patch features p_t and its nearest normal adapted features coreset c^* from memory bank $\mathcal{M}$

$$s'_t = \min_{c^* \in \mathcal{M}} E(p_t, c^*) \quad (8)$$

$$s_t = \left(1 - \frac{\exp(s'_t)}{\sum_{c' \in \mathcal{N}_b(c^*)} \exp(E(p_t, c'))} \right) s'_t \quad (9)$$

where $E(\cdot)$ is the Euclidean distance and $\mathcal{N}_b$ is the b -nearest neighbor coresets of c^* in the memory bank. Finally, the image-level anomaly score is calculated by the maximum anomaly score for all patches in the image.

IV. EXPERIMENTS

A. Experimental Settings

1) *Datasets*: We use the MVTec AD [13], VisA [45], and BTAD [46] datasets for our experiments.

MVTec AD consists of 15 categories including five types of texture and ten types of object, and comprises 3629 training images and 1725 test images. VisA contains 12 subsets with a total of 10821 images, of which 9621 are normal images and 1200 are anomalous images. The dataset is comprised of three categories: single-object images, multiobject images, and printed circuit boards. BTAD consists of three industrial products, ranging in size from 600×600 to 1600×1600 pixels. It includes 1799 images and 1031 images for training and testing, respectively.

2) *Evaluation Metrics*: Since determining of normal and anomalous samples is regarded as a binary classification problem, we use the area under the receiver operating characteristic curve (AUROC) metric to evaluate the anomaly detection performance. Following the previous studies [4], [5], we calculate the image-level and pixel-level AUROC scores for detection and segmentation performance, respectively.

3) *Implementation Details*: We used a WideResNet50 [27] pretrained on ImageNet [18] as the feature extractor. An inverse WideResNet50 is utilized as the discriminative network, operated with up-sampling, concatenating, and dimension compression operations. For the contrastive learning network, we performed adaptive average pooling operations on the feature maps of each layer and then fed them to the projector and predictor, respectively. The projector and predictor consist of a three-layer and one-layer MLP, respectively. During training, we removed the gradient of only the projector branch, and the projectors of both branches shared the weights. For the contrastive learning network, we used the same approach as [36] to construct two cropped images as positive samples. In building the memory bank, we employed a coreset subsampling percentage of 0.1 to obtain a compact one.

In the discriminative network pretraining (DNP) training phase, we used the Adam optimizer with an initial learning rate of 0.0001. In the NCL training phase, we used the stochastic gradient descent optimizer with a weight decay of 0.0001 and a momentum of 0.9, and employed the cosine annealing algorithm to adjust the learning rate. For all classes of the MVTec AD dataset and the VisA dataset, we trained 100 epochs in the first stage and another 100 epochs in the second stage. For the BTAD dataset, we trained 150 epochs in the second stage.

B. Experiment Results and Analysis

1) *Performances on MVTec AD*: We evaluated the anomaly detection performance of our proposed model compared to

TABLE I
IMAGE-LEVEL AND PIXEL-LEVEL AUROC (%) ON MVTEC AD DATASET

	SPADE [47]	PaDiM [5]	PatchCore [4]	FAPM [32]	RD4AD [33]	CFLOW-AD [6]	DeSTSeg [34]	PyramidFlow [25]	MMR [48]	ToCoAD + CutPaste [16]	ToCoAD + Perlin [31]
Bottle	98.1 / 98.4	- / 98.3	100 / 98.6	100 / 98.2	98.7 / 96.6	99.8 / 98.5	- / 99.2	- / 95.9	100 / 98.3	100 / 98.4	100 / 98.5
Cable	93.2 / 97.1	- / 96.7	99.5 / 98.4	99.2 / 98.5	97.4 / 91.0	97.6 / 97.6	- / 97.3	- / 92.1	97.8 / 95.4	98.9 / 98.2	98.7 / 98.3
Capsule	98.2 / 99.0	- / 98.5	98.1 / 98.8	98.6 / 99.0	98.7 / 95.8	97.6 / 98.5	- / 99.1	- / 96.1	96.9 / 98.0	98.6 / 98.9	98.7 / 99.0
Carpet	98.5 / 97.5	- / 99.1	98.4 / 99.0	99.0 / 98.9	98.9 / 98.9	98.5 / 98.7	- / 96.1	- / 90.8	99.6 / 98.8	97.9 / 98.8	98.0 / 98.7
Grid	99.0 / 93.7	- / 97.3	98.2 / 98.7	98.0 / 97.8	99.3 / 97.6	99.3 / 98.4	- / 99.1	- / 94.2	100 / 99.0	98.5 / 98.4	98.7 / 98.5
Hazelnut	98.7 / 99.1	- / 98.1	100 / 98.7	99.9 / 98.6	98.9 / 95.5	99.8 / 98.6	- / 99.6	- / 98.0	100 / 98.5	100 / 98.3	100 / 98.4
Leather	99.1 / 97.5	- / 99.2	100 / 99.3	99.7 / 99.0	99.4 / 99.1	99.8 / 99.0	- / 99.5	- / 99.6	100 / 99.2	100 / 99.1	100 / 99.2
Metal_nut	96.7 / 98.1	- / 97.0	100 / 98.4	100 / 98.0	97.3 / 92.3	99.2 / 98.6	- / 98.6	- / 92.8	99.9 / 95.9	99.9 / 98.5	100 / 98.4
Pill	96.5 / 96.5	- / 95.7	96.2 / 97.4	96.0 / 98.0	98.2 / 96.4	96.8 / 98.8	- / 98.7	- / 96.2	98.2 / 98.4	94.8 / 98.3	97.0 / 98.3
Screw	99.5 / 98.7	- / 98.5	98.0 / 99.3	95.2 / 99.0	99.6 / 98.2	93.5 / 98.8	- / 98.5	- / 94.0	92.5 / 99.5	96.8 / 99.5	97.0 / 99.5
Tile	89.8 / 87.4	- / 94.1	98.4 / 95.6	99.4 / 95.2	95.6 / 90.6	99.2 / 98.0	- / 98.0	- / 97.9	98.7 / 95.6	99.2 / 97.7	99.5 / 97.5
Toothbrush	98.9 / 97.9	- / 98.5	99.7 / 98.7	100 / 98.5	99.1 / 94.5	99.6 / 98.5	- / 99.3	- / 98.9	100 / 98.4	100 / 98.7	100 / 98.7
Transistor	81.0 / 94.1	- / 97.5	100 / 96.3	100 / 98.2	92.5 / 78.0	95.1 / 95.1	- / 89.0	- / 97.4	95.1 / 90.2	98.4 / 94.5	99.8 / 95.5
Wood	95.0 / 88.4	- / 94.9	99.2 / 95.0	99.1 / 94.0	95.3 / 90.9	99.0 / 95.6	- / 97.7	- / 93.8	99.1 / 94.8	99.8 / 95.8	99.6 / 95.7
Zipper	98.8 / 96.5	- / 98.3	99.4 / 98.8	99.3 / 98.6	98.2 / 95.4	98.5 / 98.7	- / 99.1	- / 95.4	97.6 / 98.0	99.4 / 98.9	99.5 / 98.9
Texture avg.	96.28 / 92.90	- / 96.92	98.84 / 97.52	99.04 / 96.98	97.36 / 95.80	99.16 / 97.94	- / 98.08	- / 95.26	99.48 / 97.48	99.08 / 97.96	99.16 / 97.92
Object avg.	95.96 / 97.54	- / 97.71	99.09 / 98.34	98.82 / 98.46	99.01 / 98.81	97.75 / 98.17	- / 97.84	- / 95.68	97.80 / 97.06	98.69 / 98.22	99.07 / 98.35
Total avg.	96.06 / 95.99	- / 97.44	99.01 / 98.07	98.89 / 97.96	98.46 / 97.80	98.22 / 98.09	- / 97.92	- / 95.54	98.36 / 97.20	98.82 / 98.13	99.10 / 98.21

TABLE II
IMAGE-LEVEL AND PIXEL-LEVEL AUROC (%) ON VISA DATASET

	SPADE [47]	DRAEM [14]	PatchCore [4]	FastFlow [35]	CFLOW-AD [6]	ToCoAD + CutPaste [16]	ToCoAD + Perlin [31]
Candle	91.0 / 97.9	91.8 / 96.6	99.1 / 98.8	92.4 / 94.2	93.5 / 98.5	96.4 / 98.9	96.4 / 98.9
Capsules	61.4 / 60.7	74.7 / 98.5	75.1 / 99.1	71.2 / 75.3	63.2 / 94.7	82.9 / 99.4	84.8 / 99.5
Cashew	97.8 / 86.4	95.1 / 83.5	97.2 / 99.1	90.7 / 91.2	94.8 / 99.1	96.5 / 98.6	96.9 / 98.5
Chewinggum	85.8 / 98.6	94.8 / 96.8	99.0 / 98.9	91.4 / 98.6	99.1 / 98.5	98.1 / 98.9	98.2 / 98.9
Fryum	88.6 / 96.7	97.4 / 87.2	95.7 / 92.7	88.6 / 97.3	93.1 / 95.9	95.8 / 92.0	96.5 / 92.1
Macaroni1	95.2 / 96.2	97.2 / 99.9	97.4 / 99.3	98.3 / 97.3	88.2 / 98.7	98.3 / 99.7	98.5 / 99.7
Macaroni2	87.9 / 87.5	85.0 / 99.2	76.4 / 98.5	86.3 / 89.2	66.5 / 96.7	81.4 / 99.1	81.1 / 99.2
PCB1	72.1 / 66.9	47.6 / 88.7	98.0 / 99.5	77.4 / 75.2	97.0 / 99.1	98.7 / 99.8	98.8 / 99.8
PCB2	50.7 / 71.1	89.8 / 91.3	97.5 / 98.7	62.2 / 67.3	89.4 / 96.6	96.4 / 98.3	96.6 / 98.3
PCB3	90.5 / 95.1	92.0 / 98.0	98.2 / 98.1	74.3 / 94.8	97.9 / 83.2	97.7 / 99.3	97.8 / 99.3
PCB4	83.1 / 89.0	98.6 / 96.8	99.5 / 98.2	80.9 / 89.9	98.6 / 98.1	99.7 / 98.1	99.8 / 98.2
Pipe fryum	81.1 / 81.8	99.8 / 85.8	99.7 / 98.7	72.2 / 87.3	99.1 / 99.3	98.8 / 98.7	98.8 / 98.7
Total avg.	82.1 / 85.65	88.65 / 93.52	94.40 / 98.25	81.25 / 88.13	90.05 / 96.56	95.06 / 98.40	95.35 / 98.43

advanced methods [4], [5], [6], [25], [32], [33], [34], [47], [48].

As shown in Table I, our models achieved the best results for at least seven categories in image-level anomaly detection, with five of them reaching 100%, and produced the best AUROC scores of 99.10% at the image-level and 98.21% at the pixel-level compared to advanced methods. However, the image-level results for some categories, such as cable, carpet, grid, pill, and screw, were nearly 1% below the best results. This could be attributed to that only the features at different local regions of the image are used for fine-tuning, which makes the model less sensitive to subtle color variations in the anomaly image. By leveraging local information, the model is better able to capture important, detailed features that are crucial for identifying anomalies. While contrastive learning may reduce sensitivity to subtle color variations, this tradeoff allows the model to focus on more significant irregularities, such as structural and textural anomalies, which are more common in real-world scenarios. This approach enhances the model's ability to detect and analyze domain-specific defects, ultimately improving overall performance. Regarding pixel-level anomaly localization performance, our method achieved optimal performance for the tile category and was within 0.7% of the best performance across almost all categories. In terms of average metrics, our model achieved

the top three for both textures and objects by narrowing the domain and allowing for the extraction of anomalous discriminative features. We visualized some representative samples for anomaly localization in the upper two rows of Fig. 5, where we can observe that our models can accurately localize anomalous regions on samples. We further presented quantitative visualization results of our approach compared to existing methods, as illustrated in Fig. 6(a). For example, in the capsule and pill examples, our model accurately localized the anomalies, whereas DeSTSeg [34] and other methods failed. It demonstrates that our method exhibits higher sensitivity to anomalous regions while effectively reducing background interference.

2) *Performances on VisA*: We evaluated our proposed method and the existing methods [4], [6], [14], [35], [47] on the VisA dataset, as illustrated in Table II. Our model achieved 95.06% image-level AUROC score and 98.40% pixel-level AUROC score based on CutPaste, while it achieved 95.35% image-level AUROC score and 98.43% pixel-level AUROC score based on Perlin noise. In particular, our model produced the highest image-level and pixel-level AUROC scores in three categories of VisA (Capsules, PCB1, and PCB4), due to the extraction of adaptive features. However, the results for Macaroni2 were lower compared to advanced methods. This discrepancy is primarily due to the fact that only some regions

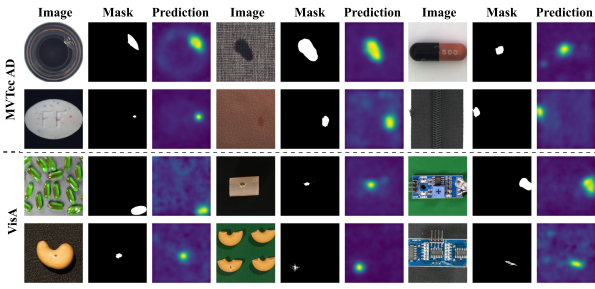


Fig. 5. Visualizations of MVTEC AD and VisA. The upper two rows are derived from the MVTEC AD, whereas the lower two rows are derived from the VisA.

TABLE III
PIXEL-LEVEL AUROC (%) ON BTAD DATASET

	01	02	03	Avg.
SPADE [47]	97.3	94.4	99.1	96.93
PaDiM [5]	96.6	95.9	98.6	97.03
PatchCore [4]	95.5	94.7	99.3	96.50
VT-ADL [46]	99.0	94.0	77.0	90.00
FYD [49]	96.1	95.3	99.7	97.03
SPR [50]	96.7	96.2	95.2	96.03
Ours + NSA [15]	97.0	96.4	99.1	97.50
Ours + CutPaste [16]	97.0	96.7	99.1	97.60
Ours + Perlin [31]	97.4	96.6	99.1	97.70

were used for fine-tuning but the global positional features of regions within the entire image are missed. This limitation led to the misclassification of positional variations as anomalies. In the lower two rows of Fig. 5, we additionally visualized some typical samples of VisA.

3) *Performances on BTAD*: We calculated the pixel-level AUROC scores of the three categories and then obtained the average score. We compared our proposed model with existing methods [4], [5], [46], [47], [49], [50], as shown in Table III. While using Perlin noise as the anomaly generator, our model achieved the best 97.70% in average pixel-level AUROC score.

4) *Ablation Study*: We explored the optimal configuration of our proposed method. We illustrated the anomaly generator we used in our experiments.

a) *Contrastive learning network*: This study was performed to determine the optimal hierarchy levels for the contrastive learning network. Table IV showed the results of ablation studies with different hierarchy levels when using CutPaste as an anomaly generator. We noticed that the model using the features extracted from layers 3 and 4 achieved the best performance among all the results, which means that the positive sample comparison learning process requires rich semantic information. Using only layer 4 lacked sufficient detail, while combining layers 2 and 3, without layer 4, produced lower results due to insufficient high-level semantics. In summary, rich deep semantic information is crucial for representing semantic features effectively.

b) *Two-stage training strategy*: We conducted experiments to investigate the role of DNP and NCL while using Perlin noise as an anomaly generator, with results shown in Table V. When only NCL was applied, the model trained solely on augmented normal samples performed poorly due to the absence of explicit negative sample guidance. Combining both DNP and NCL, the model leveraged both negative and positive samples to fine-tune the feature extractor, leading to

TABLE IV
ABLATION STUDY RESULTS ON MVTEC AD, USING DIFFERENT HIERARCHY LEVELS IN NCL TRAINING PHASE

	I-AUROC (%)	P-AUROC (%)
Layer 4	98.20	97.94
Layer 2+3	98.16	97.98
Layer 2+4	98.16	97.95
Layer 2+3+4	98.08	97.92
Layer 3+4	98.82	98.13

TABLE V
ABLATION STUDY RESULTS ON MVTEC AD, USING ONLY NCL (CL-ONLY), WITH BOTH DNP AND NCL

NCL	DNP	I-AUROC (%)	P-AUROC (%)
✗	✗	98.90	98.02
✓	✗	98.11	97.94
✓	✓	99.10	98.21

TABLE VI
ABLATION STUDY RESULTS OF DIFFERENT LOSS FUNCTION SETTINGS

λ	CE	Focal	I-AUROC (%)	P-AUROC (%)
0.1	✓		98.12	97.98
		✓	98.15	97.98
0.2	✓		98.19	97.98
		✓	98.04	97.96
0.5	✓		98.15	97.95
		✓	98.82	98.13
0.8	✓		98.18	97.94
		✓	98.13	97.94
0.9	✓		98.02	97.96
		✓	98.30	97.95

better pixel-level localization. As shown in Fig. 7, anomalous regions were more clearly highlighted when both methods were used, as the feature extractor learned more robust, generalized representations and reduced domain gap.

c) *Loss function*: This study was conducted to explore the effect of different settings on the loss function while utilizing CutPaste as an anomaly generator. Table VI reported the results of using cross-entropy (CE) loss and focal loss for different values of the hyperparameter λ . We observed that, when λ is set to 0.5, a balance was achieved between the positive sample learning process and the negative sample bootstrap learning process, and the model using focal loss outperformed the counterpart when using CE loss. It can be demonstrated that focal loss can fully consider the imbalance between normal and anomalous samples. The hyperparametric ablation results when using focal loss as L_{dis} are depicted in Fig. 6(b). When $\alpha = 0.5$ and $\gamma = 2$, the best image-level and pixel-level AUROCs were 98.82% and 98.13%, as this configuration provides an improved balance between normal and anomalous samples. When α and γ are set to different values, the effect showed minor fluctuations due to the focus on anomalous samples.

d) *Synthetic anomaly strategies*: We compared three common methods as the anomaly generator: random patches [16], random seamless patches [15], and Perlin noise [31]. The first two methods produce anomalies that are closer to the normal sample distribution, while the latter generates anomalies with greater diversity in shapes. As shown in Table VII, our methods incorporating Perlin noise achieved the best AUROCs of 99.10% at the image-level and 98.21% at the pixel-level. We considered that it is mainly attributed to the diversity of the shape and distribution of the anomalies

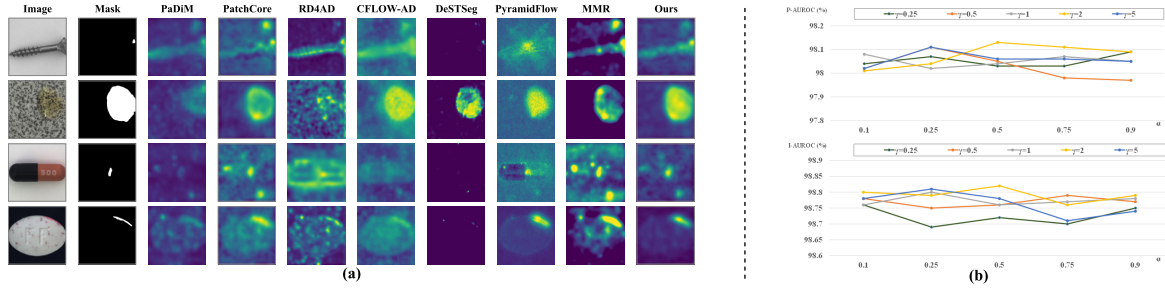


Fig. 6. (a) Anomaly samples with heatmaps on MVTEC AD. The first two columns show the defect image and the corresponding mask, and the other columns are the methods' heatmaps. (b) Visualizations of different hyperparameters of L_{dis} on MVTEC AD. The horizontal and vertical axis represents α and performance, respectively, and the lines in various colors denote different values of γ .

TABLE VII
ABLATION STUDY RESULTS ON MVTEC AD USING
DIFFERENT ANOMALY GENERATORS

	Object avg.	Texture avg.	Total avg.
CutPaste [16]	98.93 / 98.31	99.00 / 97.92	98.82 / 98.13
NSA [15]	97.75 / 98.04	98.64 / 97.82	98.04 / 97.96
Perlin [31]	99.07 / 98.35	99.16 / 97.92	99.10 / 98.21

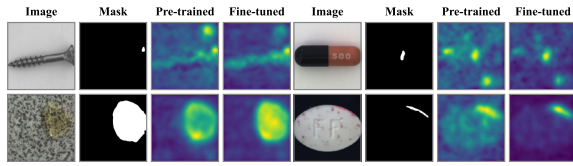


Fig. 7. Visualizations of results with the two-stage training strategy. From left to right, the original image, the ground truth, the result without two-stage training (pre-trained), and the result with two-stage training (fine-tuned).

TABLE VIII
ABLATION STUDY RESULTS ON MVTEC AD USING DIFFERENT
FEATURE EXTRACTION NETWORKS

	I-AUROC (%)	P-AUROC (%)
ResNet50 [26]	98.67	98.01
WideResNet101 [27]	99.04	98.16
WideResNet50 [27]	99.10	98.21

generated by Perlin noise, enhancing the coarse localization and bootstrap learning ability of the discriminative network.

e) *Feature extraction network*: We investigated the impact of different feature extraction networks on model performance, using Perlin noise as an anomaly generator, as shown in Table VIII. The best image-level and pixel-level AUROCs were 99.10% and 98.21% while using WideResNet50. The performance of WideResNet101 [27] is comparable to that of WideResNet50, but it requires more computational resources. When employing ResNet50 [26] instead of WideResNet50, there was a slight decline in performance due to the simpler network architecture, but the discrepancy from the optimal outcome remains relatively modest. Moreover, the smaller network requires fewer computational resources, thereby enhancing the feasibility of deployment on compact devices.

f) *Discussion*: The proposed approach uses a two-stage training strategy, enabling efficient fine-tuning with minimal resource usage and negligible performance loss when using

smaller feature extractors. However, it has some limitations. First, to fully leverage the feature representation capability of the pretrained network, it requires two separate training sessions, extending the overall training time. Second, inference relies on memory bank feature matching, which slows down as the number of stored features increases, making it less suitable for real-time tasks. As shown in Figs. 5 and 6(a), our method remains susceptible to background interference, exhibiting subtle yellow regions appearing on the heat map. To address these limitations, future research could prioritize reducing resource consumption and lightweighting models. First, eliminating redundant background features can significantly decrease training time and storage requirements, as different sample backgrounds contribute minimally to defect detection. Second, leveraging smaller models through techniques, like lightweighting or distillation, helps mitigate training costs. Finally, adopting incremental learning enables efficient updates using small data batches as new information in industrial production.

V. CONCLUSION

To bridge the domain gap between the pretrained and target-specific features for industrial anomaly detection tasks, we proposed a novel two-stage training strategy, named ToCoAD. In the first stage, a discriminative network is trained to coarsely localize anomalies. In the second stage, the pretrained discriminative network is used to provide negative-guided information, and the contrastive learning network along with the feature extractor are jointly fine-tuned. Extensive experiments were conducted to verify the superior performance of our proposed models, achieving 99.10% image-level AUROC and 98.21% pixel-level AUROC on the MVTEC AD dataset, 95.35% image-level AUROC and 98.43% pixel-level AUROC scores on the VisA dataset, and 97.70% pixel-level AUROC on the BTAD dataset.

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